Table 9.4: Anatomical differences between dicot stem and monocot stem			
S.No.	**Characters**	**Dicot Stem**	**Monocot Stem**
1.	Hypodermis	Collenchymatous	Sclerenchymatous
2.	Ground tissue	Differentiated into cortex, endodermis and pericycle and pith	Not differentiated, but it is a continuous mass of parenchyma.
3.	Starch Sheath	Present	Absent
4.	Medullary rays	Present	Absent
5.	Vascular bundles	(a) Collateral and open	(a) Collateral and closed
		(b) Arranged in a ring	(b) Scattered in ground tissue
		(c) Secondary growth occurs	(c) Secondary growth usually does not occur.

bundles are numerous, small and closely arranged in the peripheral portion. Towards the centre, the bundles are comparatively large in size and loosely arranged. Vascular bundles are **skull or oval shaped.**

Phloem

The phloem in the monocot stem consists of sieve tubes and companion cells. Phloem parenchyma and phloem fibres are absent. It can be distinguished into an outer crushed protophloem and an inner metaphloem.

Xylem

Xylem vessels are arranged in the form of **'Y'** the two metaxylem vessels are located at the upper two arms and one or two protoxylem vessels at the base. In a mature bundle, the lowest protoxylem disintegrates and forms a cavity known as **protoxylem lacuna**.

Anatomy of a Dicot Leaf-sunflower Leaf

Internal structure of dictoyledonous leaves reveal epidermis, Mesophyll and vascular tissues.

Epidermis

This leaf is generally **dorsiventral.** It has upper and lower epidermis. The epidermis is usually made up of a single layer of cells that are closely packed. The cuticle on the upper epidermis is thicker than that of lower epidermis. The minute openings found on the epidermis are called **stomata**. Stomata are more in number on the lower epidermis than on the upper epidermis. A stomata is surrounded by a pair of **bean shaped** cells called guard cells.

Each stoma internally opens into an air chamber. These guard cells contain chloroplasts, whereas other epidermal cells do not contain chloroplasts. The main function of the epidermis is to give protection to the inner tissue called **mesosphyll**. The cuticle helps to check transpiration. Stomata are used for transpiration and gas exchange.

Mesophyll

The entire tissue between the upper and lower epidermis is called the **mesophyll (GK meso = in the middle, phyllome = leaf).** There are two regions in the mesophyll. They are **palisade parenchyma and spongy parenchyma.** Palisade parenchyma cells are seen beneath the upper epidermis. It consists of vertically elongated cylindrical cells in one or more layers. These cells are compactly arranged and are generally without intercellular spaces. Palisade parenchyma cells contain more chloroplasts than the spongy parenchyma cells. The function of palisade parenchyma is **photosynthesis.** Spongy parenchyma lies below the palisade parenchyma. Spongy cells are irregularly shaped. These cells are very loosely arranged with numerous airspaces. As compared to palisade cells,

XIth_Botany_Chapter_9_EM.indd 214 5/17/2020 9:16:40 PM